Seller Mobile App
Alpha Test Reviewer

Plain words. What each part does, the rules behind it, and what to expect when you test.

The app in one minute

Agrivive lets market sellers in Davao City post leftover vegetables so buyers can reserve them online. The buyer picks it up at the stall and pays there. The seller uses this phone app to run their stall: **add produce, see reservations, scan the buyer's QR code at pickup, chat with buyers, and promote what is left.**

Step	What happens
1. Seller posts	The seller adds a vegetable with amount, price and a photo. Buyers see it on the website.
2. Buyer reserves	The buyer reserves on the website. The produce is held for **24 hours**. No online payment.
3. Buyer comes	The buyer shows a QR code (the pickup pass) at the stall.
4. Seller scans	The seller scans the code in this app, checks the amount, takes the cash, and taps Finish Pickup.
5. Not picked up	If the buyer never comes, the reservation expires after 24 hours and the stock goes back.

Who this app is for: sellers only (Android). Buyers use the website. The app needs an internet connection.

Where things are

The bottom bar has five buttons. Everything else opens from them.

Button	What it is	In one line
Home	Your stall at a glance	Stock numbers, quick buttons, recent changes.
Reservations	Buyer reservations	See who is coming, scan pickup codes, solve problems.
Listings	Your produce	Add, change, add more, take away, archive.
Messages	Chats and questions	Buyers write to you. You answer.
Boost	Free promotion	Share your produce on Facebook and Instagram.

Other places: tap your picture on Home to open your **Profile**. From Home's **More Tools** you reach **Reports** (sales), **Promote**, **Change History** and **Weather & Demand**. Also inside the app: Reviews, Notifications and Trust (your account standing).

The parts of the app, one by one

1. Sign up, sign in and your profile

What it is. How a seller gets into the app and becomes a trusted seller.

What the seller does

- Register with email and password, or continue with Google. Agree to the Terms of Use and Privacy Policy (a checkbox).
- Type the 6-digit code sent to the email (email check).
- Fill in the seller profile: name, shop name, phone number, stall address, and drop a **pin on the map** where the stall is.
- Sign in later with password, or with a one-time code sent to the email. Optional extra safety: a 6-digit code from an authenticator app (2-step sign-in), with backup codes.
- Forgot the password? Use **Forgot Password**. Locked out because of a pause or block? Use **Restore Account**.

Rules to remember

- The **Verified account** badge means: email checked and profile completed. Only verified sellers can sell and boost.
- The map pin matters. It shows buyers where the stall is and gives your weather.
- A hidden Google reCAPTCHA check runs in the background on sign in and sign up. It is invisible, so testers will not see a puzzle.

If they ask: Why a map pin? So buyers find the stall and so the app can show local weather.

2. Home

What it is. A quick look at your stall when you open the app.

What the seller does

- Greeting and your picture (opens Profile).
- **Your Stock Now**: *Ready to Sell* (what buyers can reserve), *Selling Now* (how many produce are on sale), *Running Low*, *Out of Stock*.
- Two big buttons: **Add Vegetables** and **Scan Pickup QR**.
- **More Tools**: Reports, Promote, Change History, Weather & Demand.
- **Running Low** list with an **Add More** button, and **Recent Changes** (what you added or took away).

Rules to remember

- Numbers update when you change stock.

3. Listings (your produce)

What it is. Where the seller adds and manages vegetables for sale.

What the seller does

- **Add Vegetables**: pick the vegetable from the list, choose the unit (kilo, sack or pile), type the amount and price, add a photo.
- **Add More** (put in more stock), **Take Away** (remove stock, for example it spoiled), **Change Amount** (set the exact number). You can give a reason, and **Change History** keeps a record.
- **Edit Listing** changes name, price, photo, and sets *Alert me when stock is low* (optional).
- In Edit you can also set **Drop by (PHP) every 6 hours** and a **Final price**, used by automatic price drops (see Boost).
- **Archive** hides a listing from buyers. It is kept, not deleted.
- After adding produce, the app offers to post it on Facebook and Instagram (you can say Not Now).

Rules to remember

- Buyers see the new amount right away. The unit cannot be changed after the listing is created.
- A listing with no photo cannot be boosted. Add a photo first.

If they ask: Why keep a change history? So the seller can see what happened to the stock, for example spoiled, donated or sold outside the app.

4. Reservations

What it is. Every buyer reservation, and the pickup itself.

What the seller does

- Filter: **All, Waiting, Picked up, Cancelled, Expired**.
- Open a reservation to see the buyer, the items, and the **Reserved total**.
- **Scan Pickup QR**: allow the camera, line the buyer's code inside the frame. The app checks the code before you hand anything over.
- **Check the Amount**: collect the money from the buyer (the full amount, or note what was given), then tap **Finish Pickup**. The screen says **Pickup done!**
- Cancel a reservation with a reason (for example the produce ran out). **Report a Problem** if something went wrong (QR misuse, buyer did not follow the rules, and so on). You can add up to 5 proof files.
- Buyer questions and the buyer's review show on the reservation too, with **Reply** and **Respond to review**.

Rules to remember

- A reservation lasts **24 hours**. After that it expires by itself and the stock goes back on sale.
- Payment is cash at the stall. The app does not take money.
- **Cancelling reservations counts against the seller** (see Trust). Only cancel for a real reason.

If they ask: What if the buyer's code is invalid? The app says **Invalid Code** or **Could Not Check Code**. Tap Scan Again. Never hand over produce for an invalid code.

5. Messages

What it is. Buyers write to you in two ways: a normal chat, and public questions on a product.

What the seller does

- **Chats**: send text and photos to a buyer, retry failed messages.
- **Questions** from buyers on a product page: *Waiting for You* and *Answered*. Type the answer and tap **Send Answer**. The answer also shows publicly to other buyers.
- Open the buyer chat from a reservation too.

Rules to remember

- **Answer on time.** The system watches reply times: reminder at 12 hours, warning at 24 hours, and marked non-responsive at 48 hours. Repeated misses add trust points.

6. Boost (free promotion)

What it is. Shares your produce on the Agrivive Facebook and Instagram pages to reach more buyers. It is free and not a paid ad.

What the seller does

- Boost a listing from Listings or right after adding it. Tap **Preview Post** to see the post first.
- Tabs: **Running** (boosts in progress), **Price Drops** (waiting and finished drops), **History**.
- **Boost Settings**: turn automatic price drops on or off, and set the drop amount and final price.
- **How buyers find you** (the Visibility Guide): shows how high your listings show and why.

Rules to remember

- The first post is a normal post for any available listing that has a photo. Wait about 3 hours before boosting the same produce again.
- Only **Priority Boost** listings get **automatic price drops**: 6 hours after the post, the price drops by your peso amount, then again every 6 hours, but never below your **final price**. Each drop is posted as a flash sale.
- The posting runs through a workflow tool (n8n) that writes the caption and posts to the page. If it fails, the app shows **Did not post** and you can try again.

If they ask: Why is my listing not Priority Boost? See the next section. It is about being fresh, well stocked and reliable compared with other listings.

7. Sales report (Reports)

What it is. Simple numbers about how the stall is doing.

What the seller does

- Choose the period: **7 Days, 30 Days, 90 Days**, or All.
- See what sold, income (revenue), pickup results (**Picked Up** vs **Not Picked Up**), what is left to sell, and a comparison with the period before.
- A **Tips** section gives short advice made from the seller's own numbers.

Rules to remember

- Numbers come only from real finished pickups. If there are none yet, the screen says so. It never shows made-up numbers.

8. Weather and Holidays

What it is. Helps the seller plan.

What the seller does

- Shows today's weather and the next 3 days of rain chances for the stall's map pin.
- Lists Philippine public holidays in the next 2 weeks (from the Nager.Date service) so the seller can check pickup times.

Rules to remember

- No map pin means no weather. The screen asks the seller to add a stall pin.

9. Reviews, Notifications and Trust

What it is. How buyers rate you, how the app tells you things, and how your account standing is kept fair.

What the seller does

- **Reviews**: ratings and written comments from verified buyers who finished a pickup. The seller can **Reply**.
- **Notifications**: a list of alerts (new reservation, message, boost results, price-drop notices, trust notices) with read and unread.
- **Trust**: shows account actions, verified events, notices, and a **Request correction** button if the seller thinks a record is wrong.

Rules to remember

- A report from a buyer is only a claim. It adds no points until it is checked or backed by a recorded event.
- Only the last 90 days count toward the trust score.

How buyers see your produce (the ranking)

Buyers do not see listings in random order. The app scores every live listing and shows the best first. The method is called **AHP-TOPSIS**. In plain words:

What is checked	Share	Plain meaning
Freshness	40%	How recently you posted. The score falls to zero after 12 hours.
Stock left	30%	How much of the original amount is still available.
Seller reliability	20%	Your average rating from buyers (out of 5). No ratings yet counts as middle (0.5).
Repeat selling	10%	How often you have sold the same vegetable in the last 30 days (up to 3 times).

Score	Tier name	What it means
0.60 or more	Priority Boost	Shown first. Shown in the Priority deals row. Gets the 6-hour automatic price drops.
0.40 to 0.59	Standard Exposure	Normal place in the list.
Below 0.40	Basic / Hold	Shown lower in the list.

Good to know. The score is **relative**: each listing is compared with the other live listings. With very few listings the scores bunch up, so Priority Boost may be rare in a small test. The four shares (40/30/20/10) are the team's chosen settings for the prototype, not a proven standard. The ranking runs on the server; the app only shows the result and the tier name.

Trust rules for sellers

Every recorded problem adds points. More points mean a bigger step. Only the last 90 days count.

Problem	Points
Did not answer a verified buyer question	1
Got a 1 or 2 star rating	1
Seller cancelled a reservation	2
Misleading product information (confirmed)	2
Inappropriate communication (confirmed)	2
QR code misuse (confirmed)	3, and the account is blocked at once

Total points	What happens
4 to 5	Warning notice
6 to 7	Account paused for 12 hours
8 to 9	Paused for 24 hours
10 to 11	Paused for 3 days
12 or more, or one serious violation	Account blocked

- A **pause** ends by itself. A **block** needs the seller to **Restore Account** with an email code.
- When an account is paused or blocked, the seller **cannot sign in** and is sent to the restore screen.
- Reply timers: reminder at 12 hours, warning at 24 hours, non-responsive flag at 48 hours.
- These numbers are settings the team chose for the prototype. They can be changed.

Be honest about these (known limits)

Say these before a tester finds them. They are real.

Area	What to say
Phones	Android only for now. No iPhone version yet. The app is installed from an APK file from the website's install page.
Push notifications	Set up on the server, but **not yet proven on real phones**. Our check showed no seller phone registered, so alerts are queued but nothing was delivered. Notifications inside the app list still work.
Automatic price drops and Boost alerts	Built, but the owner has not yet walked through every case on a real build. Report anything odd.
Trust warnings and pauses	Built and checked against a test database, but not yet tested end to end by the owner.
Ranking with few listings	Scores bunch up when there are only a few live listings. Priority Boost may not appear in a small test.
Internet	Needs a connection. Weather, holidays, ranking and posting all come from the server.
Sample data	Anything marked Sample data is only for demo and is not a real purchase.

Questions you might get

Why no online payment?

Pickup and payment happen at the stall. This keeps it simple for small market sellers and avoids card fees. The app records the pickup and the amount.

Who is allowed to sell?

Only verified sellers: email confirmed and profile complete with a stall pin.

How do we stop fake or no-show buyers?

Reservations expire after 24 hours, buyers who cancel or never come collect points, repeated cases pause or block them, and every pickup needs a valid QR code.

How do we stop unfair sellers?

Buyers can rate and report. Confirmed problems add points and lead to warnings, pauses or blocks. A seller can ask for a correction.

How is a listing ranked?

By freshness, stock left, rating and repeat selling (40/30/20/10), using AHP-TOPSIS. See the ranking page.

What is the free promotion?

A post on our Facebook and Instagram pages. For top listings, the price also drops every 6 hours down to the seller's final price.

Is the seller's location public?

The stall address and pin are shown to buyers so they can find the stall. Personal data such as passwords stays private.

What about two-step sign-in?

Optional. The seller can turn it on in Profile, Sign-in Safety, with an authenticator app and backup codes.

Test checklist for the alpha (in this order)

#	Try this	You should see
1	Register a new seller, check email code, finish profile with a map pin	Verified account badge
2	Add a vegetable with photo, amount and price	It appears in Listings and on the website
3	Add More, Take Away, Change Amount, open Change History	Amount changes; each change is recorded
4	Reserve that vegetable on the website as a buyer	It appears in Reservations as Waiting
5	Scan the buyer's QR pass, check the amount, Finish Pickup	Pickup done! and status Picked up
6	Ask a question as a buyer, answer it in Messages	Answer shows on the product page
7	Boost the listing, open Preview Post	Post preview; status Posting now, then Posted
8	Open Reports and Weather and Holidays	Real numbers only; weather needs a stall pin
9	Sign out, sign in with the email code	Signed in without the password
10	Turn on two-step sign-in in Profile (optional)	QR code, then 6-digit confirm, then backup codes

If something breaks: write down the screen name, what you tapped, what you expected, and what happened. A phone screenshot helps. Tell the team right after the test.

PART 2 - Test cases we did, and how to demo each one

Below are the three test sheets (N8N Marketing Boost, AHP-TOPSIS Visibility Engine, Flagging and Trust). For every case you get the test, the expected result, **your recorded result**, and a **step-by-step demo** so you can repeat it live. Results are copied from your sheets. Nothing was re-run for this PDF.

Fix these before the panel sees the sheets

1. **FTM-06** says 5 non-responses gave a **7-day** restriction. The system does not do that. Current rule: 4 to 5 points = warning, 6 to 7 = pause 12 hours, 8 to 9 = 24 hours, 10 to 11 = 3 days, 12+ = blocked. Rewrite the result, for example: 5 verified non-responses = 5 points, warning only; the 6th point starts a 12-hour pause.
2. **FTM-10** says a user over **8 points** is blocked. Blocking starts at **12 points** (or one serious violation). At 8 points the pause is 24 hours.
3. **FTM-02** has its result pasted twice (garbled). Keep one copy.
4. The AHP-TOPSIS sheet you pasted repeats the N8N rows. The real AHP cases are WSV-001 to WSV-010 (from your results file), used below. Add the date and build number, and fill Overall Result.
5. The N8N sheet is dated 10/20/26, but N8N-007 to N8N-011 describe the later rule (price drops only for Priority Boost, set after the 6-hour notice). Re-date it to the day you last ran them.

Before any demo: get this ready

Need	How
Accounts	1 verified seller on an Android build (map pin set), 2 buyer accounts on the website, 1 new seller with no reviews. Use test accounts only.
Servers	API running, **exactly one worker** (`bun run worker:dev`). The worker checks every 60 seconds, so after any change wait about a minute.
n8n	Boost workflow active, Facebook Page and Instagram connected. Keep the Facebook Page open in a browser tab.
Database	A SQL window on the same database (for example Supabase SQL editor) to move times earlier. Only change test rows.
Feed	At least 5 other live listings with different ages, stock and ratings, or scores mean nothing (TOPSIS compares listings with each other).

Time-skip tricks (so you do not wait hours or days). Replace the ids with your test rows.

```
-- find the newest boosts
select id, status, priority_tier, priority_score from seller_visibility_boosts order by created_at
desc limit 5;

-- skip the 6 hours: price drop already set -> next cut is due now
update seller_visibility_boosts set follow_up_due_at = now() - interval '1 minute' where id =
'<boost id>';

-- skip the 6 hours: settings not set yet -> 'Set Your Price Drop' notice is due now
update seller_visibility_boosts set price_drop_setup_at = now() - interval '1 minute' where id =
'<boost id>';

-- make an unanswered buyer question look 13 / 25 / 49 hours old (one step at a time)
update product_inquiries set created_at = now() - interval '13 hours' where id = '<question id>';

-- make a seller look inactive for 22 days (warning) or 31 days (block)
update "user" set last_active_at = now() - interval '22 days' where email = '<test seller email>';

-- make a reservation expire now
update orders set expires_at = now() - interval '1 minute' where id = '<order id>';
```

A. N8N Marketing Boost

Tester: Jara Joaquin T. Jara | Date on sheet: 10/20/26 | Overall result: PASSED (11 of 11)

N8N-001 | First post — Status: P

TEST	Add produce with a photo and stock. When asked "Post it on Facebook and Instagram?" tap Preview Post. Try a high-score and a low-score listing.
EXPECTED	A normal post with no score check. Facebook and Instagram each show their own status, Waiting to post to Posted, with a View link.
YOUR RESULT	Went out as a normal post, no score check. Each page showed its own status and a View link. *Remark: Checked live. High and low score both worked.*

HOW TO DEMO

Prepare: A seller on the Android build. One fresh full-stock listing (high score) and one old low-stock listing (low score).

1. App > Listings > Add Vegetables. Pick the vegetable, unit, amount, price, add a photo, tap Save Produce.
2. When the app asks "Post it on Facebook and Instagram?", tap Preview Post, then confirm the post.
3. Watch the card: Facebook and Instagram each change from Waiting to post to Posted.
4. Tap View on each to open the real post. Repeat with the low-score listing.

Show: Both listings post. No Priority check on the first post.

Say: *Anyone can make the first post. The score only decides who gets price drops later.*

N8N-002 | Score saved after the first post — Status: P

TEST	Post a listing, open its Boost screen. Do this for a score of 0.60 or higher and one below 0.60.
EXPECTED	Score and group are saved once, when the first post is saved. 0.60+ shows Priority Boost. Lower says price drops are off and why.
YOUR RESULT	Saved once at the first post. 0.60+ listing showed Priority Boost. The lower one said price drops are off and why. *Remark: The score is checked only once.*

HOW TO DEMO

Prepare: The two listings from N8N-001.

1. App > Boost tab > open each listing's boost.
2. Read the group label on the high-score listing: Priority Boost.
3. Open the low-score listing: it says price drops are off and gives the reason.
4. Optional proof, in SQL: the 'newest boosts' query above shows priority_score and priority_tier.

Show: Tier name and the 'price drops are off' reason.

Say: *AHP-TOPSIS runs one time, right when the first post is saved, and we keep that result.*

N8N-003 | Sold out or hidden listing — Status: P

TEST	Try to post a listing with 0 kg left, or one that is archived.
EXPECTED	The app says "This listing is not available to boost". Nothing is posted.
YOUR RESULT	The app refused with that message. Nothing was posted. *Remark: Same for 0 kg and archived.*

HOW TO DEMO

Prepare: A listing you can set to 0 kg, and one you archive.

1. App > Listings > open a listing > Take Away all the stock so it shows 0 left.
2. Go to Boost and try to boost it.
3. Archive another listing (Edit > Archive) and try to boost it.
4. Read the message each time.

Show: The message "This listing is not available to boost" and no new post on the Page.

Say: *The server refuses the boost, so a sold-out or hidden listing can never be promoted.*

N8N-004 | No secret key — Status: P

TEST	Open the n8n listing-details link with no key, then with a wrong key.
EXPECTED	Both times 401 "Visibility boost authentication required". No product details shown.
YOUR RESULT	Both times 401 with that message. No details shown. *Remark: Checked with no key and a wrong key.*

HOW TO DEMO

Prepare: A boost id from the 'newest boosts' query. Use a terminal.

1. Run: curl -i https://api.agrivive.tech/integrations/visibility-boosts//context (no key).
2. Run again with a wrong key: add -H "Authorization: Bearer wrong".
3. Read the status line and the message both times.

Show: HTTP 401 and the authentication message, no product data.

Say: *Only n8n holds the real secret key. Without it the server gives nothing away.*

N8N-005 | AI post text — Status: P

TEST	Post a listing and read the Facebook post.
EXPECTED	Text written by AI through n8n, shows the real price, no AI-written link. A wrong price is replaced by the backend's own text.
YOUR RESULT	Text from the AI with the real price. No link from the AI. A wrong price was replaced. *Remark: Checked on the Facebook post.*

HOW TO DEMO

Prepare: A posted listing. n8n open in another tab.

1. Open the Facebook post and read the caption.
2. Check the price in the text equals the price in the app.
3. In n8n, open the latest run and show the node 'Write Caption (Groq)', then 'Format Caption'.

Show: Caption with the correct price and no link added by the AI.

Say: *The AI writes the words. Our server checks the price and swaps the caption if the AI got it wrong.*

N8N-006 | Post fails, then retry — Status: P

TEST	Post while one or both pages refuse, then tap Post Again.
EXPECTED	The failed page shows Did not post. Post Again goes back to Waiting to post, not the old failure. Each page shows its own new result. A low score does not block the retry.
YOUR RESULT	Failed page showed Did not post. After retry: Waiting to post, then its own new result. Low score did not block. *Remark: Old failure no longer shows after a retry.*

HOW TO DEMO

Prepare: A way to make a page refuse: in n8n, temporarily disable the Facebook post node (or use a wrong Page token). Turn it back on before the retry.

1. Post a listing while the Facebook step is broken.
2. Show the Facebook line says Did not post and the card says the post did not go out.
3. Fix n8n (enable the node or restore the token).
4. Tap the Post again button on the card.
5. Show it returns to Waiting to post, then shows the new result with a View link.

Show: Did not post first, then a clean new result.

Say: *A retry starts fresh. It never shows the old error.*

N8N-007 | 6-hour notice — Status: P

TEST	Post a Priority Boost listing and wait 6 hours (skip by moving the saved time).
EXPECTED	Seller gets a "Set Your Price Drop" notice 6 hours after the first post. A listing that is not Priority Boost gets none.
YOUR RESULT	Priority seller got the notice at 6 hours. The other listing got none. *Remark: Hours skipped by moving the saved time earlier.*

HOW TO DEMO

Prepare: A Priority Boost listing posted with **no** drop amount set, and a non-Priority listing.

1. Run the 'newest boosts' query to get the Priority boost id.
2. Run the price_drop_setup_at update for that id. Wait about 1 minute for the worker.
3. App > Notifications: show 'Set Your Price Drop'.
4. Open the non-Priority listing's boost: show it got no notice.

Show: The notice only for the Priority listing.

Say: *Only top-scoring listings earn price drops, so only they are asked to set one.*

N8N-008 | Set the price drop — Status: P

TEST	On a Priority Boost cabbage at PHP 100, set 10 less each time and 70 as the final price, then save.
EXPECTED	Price drops once to 90.00 right away and is saved. Next drop is set 6 hours later. n8n never writes to the database.
YOUR RESULT	Dropped once to 90.00 right away and saved. Next drop set 6 hours later. n8n wrote nothing to the database. *Remark: Hours skipped by moving the saved time.*

HOW TO DEMO

Prepare: A Priority Boost cabbage listing at PHP 100.

1. App > Boost > Price Drops (or the notice). Enter Drop by 10 and Final price 70, tap Save Price Drop.
2. Open Listings: show the price is now 90.00.
3. Show the next drop time is 6 hours ahead.
4. Point out in n8n that it only asks the API; the database change is made by the server.

Show: PHP 90.00 and a next-drop time.

Say: *n8n only sends the request. The server does the price change, so the database stays safe.*

N8N-009 | Final price holds — Status: P

TEST	On a Priority Boost carrot at PHP 75, set 10 less each time and 70 as the final price. Let the next drop run.
EXPECTED	Price stops at exactly 70.00, never lower. If already 70.00, no cut and no second post.
YOUR RESULT	Stopped at 70.00 and never lower. At 70.00 there was no cut and no second post. *Remark: The final price held.*

HOW TO DEMO

Prepare: A Priority Boost carrot at PHP 75, drop amount 10, final price 70.

1. Save the price drop. The first cut takes 75 to 70 (not 65).
2. Run the follow_up_due_at update to trigger the next round.
3. Wait a minute. Show the price is still 70.00 and no new Facebook post appeared.

Show: Price 70.00 and no second post.

Say: *The seller's final price is a floor. We never go below it.*

N8N-010 | Second post — Status: P

TEST	After the cabbage drops to 90, check the second Facebook post.
EXPECTED	The second post shows PHP 90 and is saved as the second post for that listing.
YOUR RESULT	Showed 90 and was saved as the second post. *Remark: Checked on the Facebook Page.*

HOW TO DEMO

Prepare: The cabbage from N8N-008.

1. Open the Facebook Page and find the newest cabbage post.
2. Read the price in the post: 90.
3. App > Boost > History: the listing shows the price-drop post.

Show: A flash-sale post with the new price.

Say: *Every price drop is announced to buyers with the new price.*

N8N-011 | Price drop only for Priority Boost — Status: P

TEST	On a listing below 0.60 at its first post, tap Lower Price Now or save a price drop.
EXPECTED	App says "Price drops need Priority Boost". Price does not change, no second post.
YOUR RESULT	Showed that message. Price stayed, no second post. *Remark: Same for Lower Price Now and for saving a drop.*

HOW TO DEMO

Prepare: A non-Priority listing that has been posted.

1. App > Boost > Price Drops tab: tap Lower Price Now on the non-Priority listing.
2. Read the message.
3. Try to save a drop amount and final price in Boost Settings and read the message again.
4. Show the price in Listings did not change.

Show: The refusal message and an unchanged price.

Say: *Price drops are a reward for top listings only.*

B. AHP-TOPSIS Multi-Criteria Decision Visibility Engine

Tester: Jara Joaquin T. Jara | Test date: November 1, 2026 | Result: 10 of 10 passed

Prepare once for all WSV cases: set `VISIBILITY_POLICY=visibility-ahptopsis-v1` and restart the API. Have at least 5 other live listings with different age, stock and ratings. Remember: weights are freshness 0.40, stock 0.30, reliability 0.20, repeat selling 0.10. Tiers: 0.60 and up Priority Boost, 0.40 to 0.59 Standard Exposure, below 0.40 Basic/Hold. A listing alone in the feed always scores 0.50.
How to see scores live: open `https://api.agrivive.tech/marketplace/products` (or the OpenAPI page) and read `visibilityScore`, `visibilityTier` and `visibilityPolicyVersion` on each item.

WSV-001 | Fresh tomatoes, full stock — Status: P

TEST	Publish a brand-new tomato listing (10 kg, PHP 40/kg) with full stock, photo and pickup location from an active, well-rated seller.
EXPECTED	Priority Boost (score 0.60 or higher) and shown at the top of the feed with a Priority badge.
YOUR RESULT	Priority Boost (score 0.78), ranked first. *Remark: Checked with 5 other listings. The score depends on the others.*

HOW TO DEMO

Prepare: The seeded feed. A well-rated seller account.

1. Seller app: add the tomato listing with photo.
2. Open the website marketplace and refresh.
3. Show it at the top.
4. Open the API list and show visibilityTier = priority and the score.

Show: Top position, Priority tier, score 0.60+.

Say: *Fresh and fully stocked beats everything else, so it goes first.*

WSV-002 | Moderate age, half stock — Status: P

TEST	Cabbage published 6 hours ago with 5 of 10 kg left, average-rated seller.
EXPECTED	Standard Exposure (0.40 to 0.59), in the middle of the feed.
YOUR RESULT	Standard Exposure (0.50), 3rd of 7. *Remark: Same feed as WSV-001.*

HOW TO DEMO

Prepare: A cabbage listing. You cannot make a listing 'older' from the app, so use SQL on the test listing.

1. Run: update sellers_product set published_at = now() - interval '6 hours', product_qty = 5 where id = ''; (original amount stays 10)
2. Refresh the marketplace.
3. Show the cabbage in the middle and read its tier from the API.

Show: Standard tier, middle position.

Say: *Half the stock and 6 hours old puts it in the middle.*

WSV-003 | Old, low stock — Status: P

TEST	Eggplant published over 12 hours ago with 1 kg left of 20 kg, unrated or low-rated seller.
EXPECTED	Basic/Hold (below 0.40), near the bottom.
YOUR RESULT	Basic/Hold (0.00), at the bottom. *Remark: The lowest listing in a feed always scores 0.00.*

HOW TO DEMO

Prepare: An eggplant test listing.

1. Run: update sellers_product set published_at = now() - interval '13 hours', product_qty = 1 where id = ''; (original amount 20)
2. Refresh the marketplace.
3. Show it at the bottom and read its tier.

Show: Basic/Hold tier, last place.

Say: *Old and almost gone means it is shown last.*

WSV-004 | New seller, no reviews — Status: P

TEST	A newly registered seller with 0 reviews publishes a fresh 15 kg sitaw listing.
EXPECTED	Published without errors. Seller reliability is the neutral 50%. Ranked by freshness and stock.
YOUR RESULT	Neutral 50% used. Ranked 2nd (score 0.76). *Remark: Seller has no reviews, so neutral 50% was used.*

HOW TO DEMO

Prepare: A brand-new seller account with a finished profile.

1. Sign in as the new seller. Add a fresh 15 kg sitaw listing with photo.
2. Refresh the marketplace.
3. Show it ranks high even with no reviews, and point to reliability = 0.50.

Show: A good rank for a new seller.

Say: *A new seller is not punished for having no reviews. We assume the middle value.*

WSV-005 | Repeat seller — Status: P

TEST	Fresh pechay from a seller who listed pechay 3 or more times in 30 days, against a first-time seller.
EXPECTED	The repeat seller ranks higher.
YOUR RESULT	Repeat seller scored 0.96 against 0.75. *Remark: Counts listings in the last 30 days, not sales. Worth 10%.*

HOW TO DEMO

Prepare: Two sellers: one with 3+ earlier pechay listings in the last 30 days (prepare this ahead of time), one first-timer. Same stock and rating.

1. Post a fresh pechay listing from each seller with the same stock.
2. Refresh the marketplace.
3. Show the repeat seller listed above the first-timer and compare the two scores in the API.

Show: Two pechay listings, repeat seller first.

Say: *Sellers who keep supplying the same vegetable get a small boost, worth 10 percent.*

WSV-006 | Sold out — Status: P

TEST	Reserve the last kilograms of a squash listing so the amount becomes 0.
EXPECTED	Marked sold out with a visible badge, ordering disabled, removed from the ranking list.
YOUR RESULT	Marked sold out, badge shown, ordering disabled, removed from the ranking. *Remark: Tested live. The feed only shows stock above 0.*

HOW TO DEMO

Prepare: A squash listing with a small amount and a buyer account on the website.

1. As the buyer, reserve the full remaining amount.
2. Refresh the marketplace: the listing is gone from the feed.
3. Open the product page: it shows Sold out with ordering disabled.

Show: Missing from the feed, Sold out on the product page.

Say: *When the stock hits zero the listing leaves the ranking.*

WSV-007 | Equal scores keep a steady order — Status: P

TEST	Two listings (for example carrots and okra) posted 5 minutes apart with equal stock, rating and history.
EXPECTED	Order never swaps on refresh; the newer listing is first.
YOUR RESULT	Order stayed the same; newer first. *Remark: Ties sort newest first, then by listing id.*

HOW TO DEMO

Prepare: Two listings with the same stock from sellers with the same rating.

1. Post the two listings 5 minutes apart.
2. Refresh the marketplace 5 times.
3. Show the order never changes and the newer one stays first.

Show: A stable order.

Say: *If scores tie, the newer listing wins, so the order never jumps around.*

WSV-008 | Page by page (cursor) — Status: P

TEST	Open the marketplace with more than 20 active listings and load page 2 using the next-page marker.
EXPECTED	Each batch is in descending score order, no duplicates, no skipped items.
YOUR RESULT	Batches loaded in order with no duplicates or skips at page sizes 1, 2, 3, 5 and 7. *Remark: Live site, read only. Paging fix: commit c437e99.*

HOW TO DEMO

Prepare: A feed with 20+ listings. No new listings or stock changes during the demo.

1. Open the API list with a small limit, for example limit=5.
2. Copy nextCursor and request the next page. Repeat.
3. Check no id appears twice and the scores keep going down.

Show: Pages with no repeats.

Say: *Browsing page by page never repeats or skips a listing. A marker older than 60 minutes is refused.*

WSV-009 | Boost and the 6-hour rule — Status: P

TEST	Post a new listing on social media then check the 6-hour rule, for one Priority Boost listing and one below it.
EXPECTED	Both get a normal first post. AHP-TOPSIS runs once and the score is saved. Only Priority Boost gets the Set Your Price Drop notice after 6 hours and the drops every 6 hours.
YOUR RESULT	First post worked. The 6-hour rule worked for the Priority Boost listing. *Remark: Checked live. Hours skipped by moving the saved times.*

HOW TO DEMO

Prepare: Same as N8N-001, N8N-002 and N8N-007.

1 Follow the demo for N8N-001, then N8N-002, then N8N-007 in that order.

Show: First post for both, notice only for Priority.

Say: *The first post is for everyone. Price drops are the reward for the top tier.*

WSV-010 | Safe rollback — Status: P

TEST	Set VISIBILITY_POLICY=visibility-v4, restart, refresh the marketplace. Switch back and restart.
EXPECTED	The marketplace switches to the baseline ranking with no errors. Switching back restores AHP-TOPSIS.
YOUR RESULT	Switched to baseline with no errors. Switching back restored AHP-TOPSIS. *Remark: Only visibility-v4 gives the old formula. Any other value gives AHP-TOPSIS.*

HOW TO DEMO

Prepare: Local API with a code editor open on backend/.env.

1 Edit backend/.env: VISIBILITY_POLICY=visibility-v4. Restart the API.
2 Refresh the marketplace. In the API, visibilityPolicyVersion says visibility-v4.
3 Edit back to visibility-ahptopsis-v1. Restart. Refresh. The version is back and the order changes again.

Show: The policy version string switching both ways.

Say: *If the new ranking ever misbehaves, we can go back to the old formula with one setting.*

C. Flagging and Trust Mechanism

Tester: Jara Joaquin T. Jara | Date: not filled in | Result: 10 of 10 passed (see the fixes box on the first page of Part 2)

Current rules to quote (policy monitoring-v3): reply timers 12 hours reminder, 24 hours warning plus buyer notice, 48 hours non-responsive (1 point). Points: non-response 1, low rating 1, seller cancellation 2, buyer no-show 2, misleading info 2 (confirmed), inappropriate message 2 (confirmed), QR misuse 3 (serious, blocks at once). Steps: 4 to 5 warning, 6 to 7 pause 12 hours, 8 to 9 pause 24 hours, 10 to 11 pause 3 days, 12 or more blocked. Only the last 90 days count. A report counts only after it is confirmed (2 different people, or system proof). Inactivity: 21 days warning, 30 days block.
Where to look: the seller app Trust screen (points, notices, verified events) and Notifications.

FTM-01 | Clean account — Status: P

TEST	A seller has no confirmed violations.
EXPECTED	No warning, restriction or block. Points 0, action None.
YOUR RESULT	Points 0, action None. *Remark: Clean account resolves to no action.*

HOW TO DEMO

Prepare: A seller account with no events.

1. Sign in as the clean seller.
2. Open Profile > Trust (account standing).
3. Show 0 points, no actions, no notices.

Show: Zero points and no action.

Say: *A seller with a clean record is left alone.*

FTM-02 | Report with no proof — Status: P

TEST	A buyer reports a seller for misleading information, but nothing in the records supports it.
EXPECTED	Report is saved for review. No warning, restriction or block without evidence.
YOUR RESULT	Saved as an allegation. 0 penalty points. *Remark: Unsubstantiated reports do not affect standing.*

HOW TO DEMO

Prepare: A completed order between a buyer and the seller.

1. As the buyer on the website: Orders > open the order > Report an issue with this order.
2. Choose misleading product information, write a reason, submit with no files.
3. As the seller: Trust screen. The report appears under Unverified reports.
4. Show points are still 0. The screen says 'Reports add no points.'

Show: Report listed, 0 points.

Say: *A report is only a claim. It never punishes anyone by itself.*

FTM-03 | 12 hours, no reply — Status: P

TEST	A seller gets a buyer message and does not reply within 12 hours.
EXPECTED	Reminder sent to the seller. No warning, not marked non-responsive.
YOUR RESULT	Reminder sent at 12 hours. 0 violation events. *Remark: Seller standing unaffected.*

HOW TO DEMO

Prepare: A buyer question on one of the seller's products, left unanswered. Get its id.

1. Buyer: ask a question on the product page.
2. SQL: update product_inquiries set created_at = now() - interval '13 hours' where id = '';
3. Wait about a minute for the worker.
4. Seller app > Notifications: show the reminder.
5. Trust screen: still 0 points.

Show: A reminder and no points.

Say: *At 12 hours we only send a friendly nudge.*

FTM-04 | 24 hours, no reply — Status: P

TEST	A seller still has not replied within 24 hours.
EXPECTED	Warning to the seller and a notice to the buyer.
YOUR RESULT	Warning sent to the seller and an update notice delivered to the buyer. *Remark: Both notices sent.*

HOW TO DEMO

Prepare: The same question from FTM-03.

1. SQL: set created_at to now() - interval '25 hours' for that question. Wait about a minute.
2. Seller app > Notifications: show the warning.
3. Buyer on the website > Notifications: show the notice that the seller has not replied.

Show: Two notices, one each side.

Say: *At 24 hours the seller is warned and the buyer is told.*

FTM-05 | 48 hours, marked non-responsive — Status: P

TEST	A seller still has not replied within 48 hours.
EXPECTED	Seller marked non-responsive and the incident saved in the violation record.
YOUR RESULT	Marked non-responsive. Verified violation logged. *Remark: Committed to the violation record.*

HOW TO DEMO

Prepare: The same question from FTM-03.

1. SQL: set created_at to now() - interval '49 hours'. Wait about a minute.
2. Seller app > Trust > Verified events: show the non-response event.
3. Show the total is now 1 point.

Show: One verified event, 1 point.

Say: *After 48 hours the missed reply is written into the seller's record.*

FTM-06 | Repeated non-response adds up — Status: P

TEST	A seller has several confirmed non-response incidents.
EXPECTED	Incidents add to the weighted score. At the configured threshold a temporary restriction is applied.
YOUR RESULT	5 incidents = 5 points, then a 7-day restriction. **(Does not match the current rule: fix before presenting.)** *Remark: Correct this result to the current rule.*

HOW TO DEMO

Prepare: Prepare ahead: create 6 unanswered buyer questions on different products of a test seller, or use a seller that already has the points.

1. Set all 6 questions to 49 hours old (SQL as FTM-05) in one go. Wait about a minute.
2. Seller app > Trust: show the 4th point gives a warning and the 6th point gives a 12-hour pause.
3. Try to sign in as that seller during the pause: it is refused and sent to Restore Account.

Show: Points climbing and the pause screen.

Say: *More confirmed misses means a bigger step: warning first, then a short pause.*

FTM-07 | Three weeks inactive — Status: P

TEST	A seller does nothing in Agrivive for three weeks.
EXPECTED	Inactivity warning sent. Account stays active.
YOUR RESULT	21-day inactivity detected. Warning sent. Account active. *Remark: Warning only.*

HOW TO DEMO

Prepare: A test seller.

1. SQL: update "user" set last_active_at = now() - interval '22 days' where email = '';
2. Wait about a minute.
3. Seller app > Notifications: show the inactivity warning.
4. Show the account still works.

Show: A warning notice and a working account.

Say: *Three weeks quiet gets a warning, not a punishment.*

FTM-08 | One month inactive — Status: P

TEST	A seller is inactive for one month.
EXPECTED	Account blocked by the inactivity rule.
YOUR RESULT	30-day inactivity detected. Account set inactive (blocked). *Remark: Reactivation needs an email code.*

HOW TO DEMO

Prepare: A test seller.

1. SQL: set last_active_at to now() - interval '31 days'. Wait about a minute.
2. Try to sign in: you are sent to Restore Account.
3. Enter the email and the 6-digit code to restore.
4. Sign in again: it works and the inactivity clock restarts.

Show: A blocked account that can be restored.

Say: *A silent account is closed to keep the market honest, but the seller can come back with an email code.*

FTM-09 | Buyer no-show — Status: P

TEST	A buyer reserves vegetables but does not arrive in the allowed time.
EXPECTED	The system checks the reservation and records a confirmed no-show violation.
YOUR RESULT	Reservation expired. No-show recorded against the buyer with 2 points. *Remark: Buyer is penalised for abandoning a reservation.*

HOW TO DEMO

Prepare: A buyer account and a seller product. Get the order id.

1. Buyer reserves a product on the website.
2. SQL: update orders set expires_at = now() - interval '1 minute' where id = ''; Wait about a minute.
3. Buyer's Orders page: the order shows Expired and the stock returns to the seller.
4. Show the missed-pickup notice to the buyer. For the 2 points, show the trust event in the database or the admin trust view.

Show: Expired order and 2 points.

Say: *A buyer who never shows up ties up food, so it counts against them. Sellers are held to rules too.*

FTM-10 | Weighted score decides the action — Status: P

TEST	A user has several confirmed violations of different types.
EXPECTED	Score = type weights times confirmed events. Compared with thresholds: none, warning, pause or block.
YOUR RESULT	Score calculated with the assigned weights. A user above 8 points was blocked. **(Does not match the current rule: blocking starts at 12 points. Fix before presenting.)** *Remark: Tiered system works: warning, pause, block.*

HOW TO DEMO

Prepare: Prepared test accounts at about 4, 6, 8, 10 and 12 points, or use the policy table and one live example.

1. Show the points table (see the box above).
2. Open the Trust screen of a seller at 8 points: a 24-hour pause. At 12 points: blocked.
3. Explain: one serious violation (QR misuse, 3 points) blocks at once, but only after two different people report it or the system proves it.

Show: Points, the action, and the pause or block message.

Say: *The more confirmed problems, the stronger the step. It is gentle first and strict only for serious or repeated cases.*

References

Where each idea in this reviewer comes from. Methods and academic papers first, then services the app uses, UX laws, and project sources.

A. Ranking method (AHP and TOPSIS)

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Hwang, C.-L., & Yoon, K. (1981). *Multiple Attribute Decision Making: Methods and Applications, A State-of-the-Art Survey.* Lecture Notes in Economics and Mathematical Systems, vol. 186. Springer-Verlag, Berlin Heidelberg. https://doi.org/10.1007/978-3-642-48318-9. *Used for: vector normalisation, positive/negative ideal solutions, Euclidean distances, and relative closeness score (C_i*).*

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About our weights. The papers above support the mathematical formulation of AHP and TOPSIS. The specific criteria shares (Freshness 40%, Stock left 30%, Seller reliability 20%, Repeat selling 10%) are the Agrivive team's prototype choices for field trials. If needed, pairwise judgements from agricultural stakeholders can formally recalibrate them.

B. Trust, flagging, and marketplace governance

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OECD (2022). *The role of online marketplaces in protecting and empowering consumers.* OECD Digital Economy Papers, No. 334. OECD Publishing, Paris. https://doi.org/10.1787/9d8cc586-en. *Supports graduated sanctions: warning -> temporary restriction/pause -> suspension.*

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About our trust numbers. Academic research supports the weighted penalty architecture and progressive sanction tiers (warning, pause, block). The exact numbers (1, 2, 3 points; 4, 6, 8, 10, 12 thresholds; 12h/24h/48h reply timers; 21/30-day inactivity) are Agrivive prototype rules designed for field trials and can be tuned after beta testing.

C. UI/UX standards, cognitive ergonomics, and design laws

Nielsen, J., & Molich, R. (1990). Heuristic evaluation of user interfaces. In *Proc. SIGCHI CHI '90* (pp. 249-256). ACM. https://doi.org/10.1145/97243.97281. *Jakob's Law (familiar ecommerce patterns), Recognition Over Recall (prominent status badges), and Visibility of System Status.*

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Fitts, P. M. (1954). The information capacity of the human motor system. *J. Exp. Psychol., 47*(6), 381-391. https://doi.org/10.1037/h0055392. *Minimum 48px touch CTA height and deliberate separation of destructive controls.*

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D. Services and tools the app uses

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E. Agrivive project sources and codebase artifacts

Bartiquien, A., Jara, J. T., & Marundan, J. (2026). *Agrivive: A Surplus Vegetable Marketplace with Multicriteria Visibility Ranking and Automated Social Promotions* (Capstone Project Paper). Saved as *CAPSTONE 2 FILES - BARTIQUIEN, JARA, MARUNDAN.pdf*. Reply timers (Table 17), serious violations (Table 18), and architecture (Figures 4, 5, 6, 13).

Agrivive Project Vault Notes (Obsidian): *2026-10-11 AHP-TOPSIS Visibility Implementation Plan*; *2026-11-02 Normal First Post and Priority Price Drops*; *2026-11-02 Trust Numbers and Research Sources*; *Trust and Flagging*; *Source and Reading Guide*.

Agrivive Validation Test Sheets & Results: *WSV-AHP-TOPSIS-test-sheet.md* and *WSV-AHP-TOPSIS-test-results.md* (1 Nov 2026, cases WSV-001 to WSV-010); *N8N-Boost-test-sheet.md* (cases N8N-001 to N8N-011); Flagging and Trust Mechanism test suite (cases FTM-01 to FTM-10).

Agrivive Source Code: `backend/src/utils/ahp-topsis` (TOPSIS solver), `utils/visibility-ranking` (criteria evaluation), `utils/trust` (penalty point accumulator & sanction router), `worker.ts` (scheduled 60s cron jobs), `modules/integrations`, and `n8n/workflows/agrivive-visibility-boost.json`.

Citation Verification Complete. All entries in this reference list have been verified with complete bibliographic metadata (author initials, full article and book titles, journal names, volume and issue numbers, page spans, and official DOIs or repository links). No placeholder citations remain.